Particle Swarm Optimization

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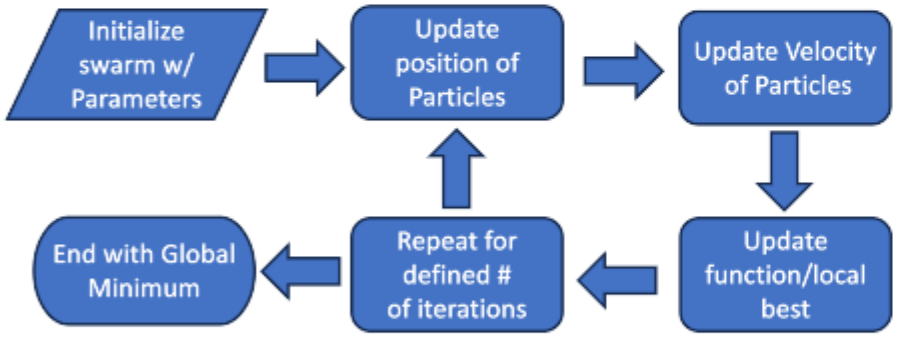
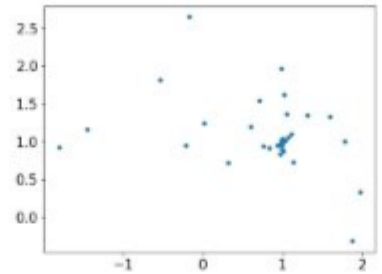
Intro To Particle Swarm Optimization

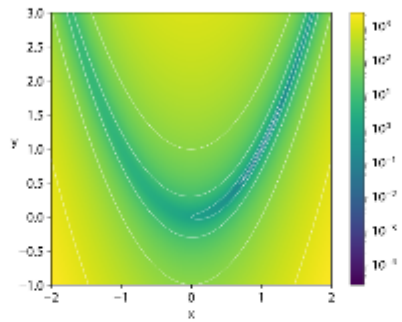
- Particle swarm optimization is an optimization method that iteratively improves candidate solutions for a cost function.
- These candidate solutions (or particles) to a problem move within the search space. Each particle dynamically adjusts its velocity based on its own position and the positions of the other particles for a set number of iterations, swarming on the global minimum.
- The solution is a heuristic solution where the best solution is never exact but is a good enough approximation.

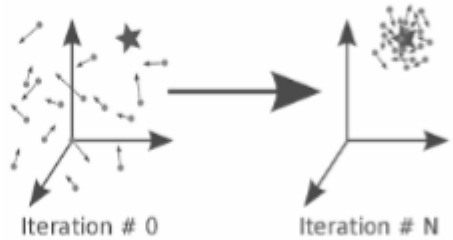


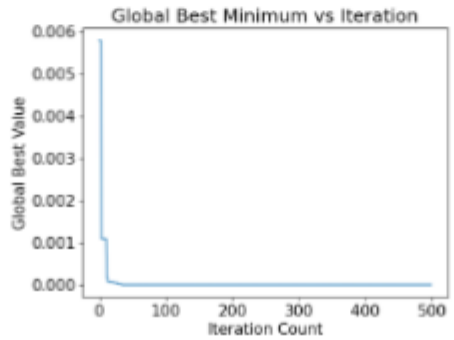
Real Life Application

- PSO is intended to simulate social behaviors such as the movement of a school of fish or flock of birds.
- It may also be used to optimize the timing of traffic signals at intersections to minimize congestion, reduce travel time, and improve overall traffic flow.
- The model would consider variables such as the duration of the green light and wait time to find the ideal timing of the traffic light changes.









Limitations

- If there are multiple local minima, it is hard to find the global minimum; particles will get stuck at local minima and never reach the global minimum.
- It is difficult to model real life interactions using PSO as more complex interactions have many parameters in the cost function, which makes the particles more complex. Our code has 2 parameters, the x and y, while a real-life problem could have hundreds.

QR Codes

Code

Visual Representation

References

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